

A Boosted Meta-Ensemble of Deep Learning Models for Short-Term Flood Forecasting in the Mahanadi River Basin

Abstract—This study presents a robust hybrid forecasting framework that fuses stacked deep learning (DL) sequence models with boosted meta-learning to produce accurate 5-day ahead river inflow forecasts. Three complementary DL branches—an LSTM with multi-head attention, a PatchTST-inspired dense compression + attention model, and a TFT-inspired LSTM+attention pathway—are first stacked to generate multi-horizon predictions. These DL outputs are then provided as features to per-horizon boosted meta-models (XGBoost, with additional out-of-fold contributions from LightGBM and Random Forest). A final blending step stabilizes forecasts by combining meta-predictions with the mean of base learners. On Mahanadi Basin data (2000–2014, 162 predictors), the proposed Boosting+Stacking fusion consistently outperformed the DL-only stacked model across all horizons. For Day-1 forecasts, Nash–Sutcliffe Efficiency (NSE) improved from 0.857 (stacking) to 0.961, with RMSE reduced from 1430.82 to 751.05 and MAE from 668.18 to 412.56. Similar gains were observed through Day-3 (NSE rising from 0.769 to 0.840, KGE from 0.648 to 0.880). Importantly, the method sustained reliable skill even at longer horizons, achieving $\text{NSE} > 0.90$ through Day-2 and maintaining NSE above 0.70 at Day-5 (0.712 vs. 0.633 for stacking). These improvements, consistently reflected across RMSE, MAE, and KGE metrics, highlight the robustness of hybrid DL+ML fusion with attention for hydrologic forecasting. By preserving predictive skill over both short and extended horizons, the framework provides actionable early-warning capabilities for operational flood forecasting and water resource management in rain-fed basins.

Index Terms—Rainfall prediction, Flood forecasting, LSTM, Ensemble learning, Multi-head attention, Mahanadi Basin

I. INTRODUCTION

Flood forecasting in large river basins has long been recognized as a critical component of disaster risk reduction and water resources management. Recent advances in machine learning (ML) and deep learning (DL) provide opportunities to design hybrid approaches that can capture complex rainfall–runoff dynamics more effectively than traditional models. This section outlines the motivation for the study, the challenges of inflow forecasting in rain-fed basins, the proposed hybrid DL–ML approach, and the novelty of this work in advancing multi-day predictive skill.

A. Motivation

Climate-related extreme weather phenomena are increasing in both frequency and intensity due to global climate change, with floods ranking among the most destructive. The Indian subcontinent, governed by a monsoon-dominated climate, is particularly vulnerable. Its extensive rain-fed river basins are subject to intense and erratic rainfall spells that frequently trigger severe flooding, leading to agricultural losses, disruption of

critical infrastructure, and threats to human life. Reliable flood forecasting systems are therefore essential, both for mitigating disaster impacts and for enabling more resilient water resource management.

Accurate inflow prediction requires models that can effectively learn the complex spatio-temporal relationships between rainfall and discharge. To achieve this, forecasting systems must integrate both short-term variability (e.g., sudden high-intensity rainfall events) and long-term hydrological memory (e.g., cumulative seasonal rainfall during monsoon periods). This study is motivated by the need to develop such a system, capable of learning rainfall–runoff dynamics at multiple time scales, while producing skillful multi-day inflow forecasts.

B. Challenges in Inflow Forecasting

Forecasting inflows in large, rain-fed basins is scientifically challenging due to the non-linear and dynamic nature of rainfall–runoff processes. River discharge is influenced by heterogeneous rainfall across sub-catchments, antecedent soil moisture, land use, and monsoon-driven variability. Process-based hydrological models, while physically interpretable, require extensive calibration and detailed physiographic data, which are often unavailable. Purely data-driven approaches such as ML and DL, though effective in approximating non-linear relationships, often face a trade-off: models optimized for short-term events may neglect seasonal patterns, whereas long-term models may overlook sudden localized floods.

C. Research Objective and Proposed Approach

To address these limitations, this study introduces a hybrid ensemble forecasting framework that integrates deep learning and machine learning through decision-level fusion. The framework begins with three complementary DL sequence models:

- An LSTM with multi-head attention, designed to capture temporal dependencies and assign variable-level importance.
- A PatchTST-inspired dense compression + attention branch, which compresses the 30-day input sequence into dense representations and applies attention (note: no Conv1D-based patch extraction is implemented).
- A TFT-inspired LSTM+attention branch, intended to fuse multi-scale seasonal and event-driven signals. Variable selection networks and gating from the full TFT are not implemented here.

The outputs of these DL branches are not used directly as forecasts but rather as input features to meta-learners. This ensures that the second-stage models can learn how best to weight and

combine the heterogeneous strengths of the DL branches. Attention mechanisms additionally facilitate the capture of spatial variability in rainfall and temporal causality in runoff response, which are critical for flood-relevant processes. These stacked features are subsequently refined using boosted meta-learners (XGBoost, complemented by out-of-fold contributions from LightGBM and Random Forest). This design introduces sample weighting and cross-validated meta-features, which improve robustness in peak-flow conditions and enhance generalization across different hydrologic regimes. A final blending step averages meta-model outputs with the mean of base predictions, stabilizing the forecasts and reducing variance.

D. Novelty and Significance of the Study

This study makes the following scientific contributions:

- **Hybrid DL–ML Fusion:** Development of a two-stage ensemble where attention-based DL forecasts serve as structured features for boosted meta-learners.
- **Peak-Event Robustness:** Incorporation of out-of-fold meta-features and sample weighting strategies that emphasize extreme flows, ensuring improved reliability during flood peaks.
- **Improved Multi-Day Skill:** Demonstrated gains in forecast accuracy on the Mahanadi Basin dataset (2000–2014), with NSE improving from 0.857 to 0.961 at Day-1 and remaining above 0.70 at Day-5, thereby exceeding typical operational reliability thresholds.
- **Operational Utility:** By maintaining $NSE > 0.90$ for the first two days and above 0.70 through five days, the system provides actionable early-warning capabilities for flood forecasting in rain-fed basins.

E. Organization of the Paper

The remainder of this paper is structured as follows: Section 3 describes the study area and dataset, including rainfall and discharge inputs. Section 4 details the methodology, covering the architecture of the deep learning branches, the stacking strategy, and the boosted meta-learning refinement. Section 5 presents the experimental results, with a focus on comparing stacked and boosted-stacked forecasts under various performance metrics. Section 6 discusses the findings, limitations of the current framework, and potential operational implications. It also concludes the paper by summarizing key insights and outlining directions for future research.

In addition, various illustrative plots and visualizations are provided to support the experimental analysis—such as comparisons between predicted and actual values on the test set—and schematic block diagrams are included to depict the hybrid framework and the internal architectures of the deep learning branches. These visual elements collectively aid in understanding the data flow, model fusion strategies, and forecast performance across different configurations.

II. STUDY AREA AND DATASET

A. Study Area

The study was conducted for the Mahanadi River Basin (MRB), one of the largest river systems in east-central India and among the most flood-prone regions of the country. The

basin extends between $19^{\circ}8'–23^{\circ}32'$ N and $80^{\circ}28'–86^{\circ}43'$ E, draining an area of approximately 141,600 km² across the states of Chhattisgarh and Odisha.

The hydro-climatic regime of the MRB is dominated by the southwest monsoon (June–September), which contributes more than 70% of the annual rainfall. This, combined with heterogeneous topography and anthropogenic interventions (e.g., the Hirakud reservoir, barrages, diversions), results in highly variable hydrological responses and frequent flood events. These complexities make the MRB an appropriate and scientifically challenging test bed for developing advanced multi-step inflow forecasting models.

B. Data Sources

A daily hydro-meteorological dataset spanning 1 January 2000 – 30 September 2014 was compiled, consisting of rainfall and discharge observations.

- **Rainfall Data:** Gridded rainfall estimates at 0.25×0.25 resolution were obtained from the India Meteorological Department (IMD). Using the `imdlib` Python library, grid points located within the MRB boundaries were extracted. Each grid cell was treated as a separate predictor feature (e.g., $11, 12, \dots$), and a basin-mean rainfall series was computed by averaging across all grids.
- **Discharge Data:** Daily river discharge observations were sourced from the Central Water Commission (CWC), Government of India, for 11 gauging stations upstream of the Hirakud reservoir (Andhiyarkhore, Bamnidhi, Basantpur, Ghatota, Kelo, Kurubhata, Paramanpur, Rajim, Rampur, Simga, Sundargarh). The station-wise values were averaged to form a representative basin inflow series, which served as the target variable for model training and evaluation.

Together, these datasets capture basin-wide rainfall variability and discharge responses, providing a reliable basis for sequence-based ML/DL forecasting.

C. Data Preprocessing and Feature Engineering

A structured preprocessing pipeline was applied to ensure data quality and suitability for modeling:

- **Feature Inclusion:** All numeric hydrometeorological variables were retained as predictors, including individual rainfall grid cells ($11, 12, \dots$), basin-mean rainfall, and basin-mean discharge. The total feature count (162 in this study) depends on the number of rainfall grid points falling within the basin mask and may vary if the spatial domain or resolution is updated.
- **Missing/Zero Handling:** Zero and missing values were replaced with a small constant (0.01) to maintain numerical stability during log-transformation and scaling.
- **Target Transformation:** The inflow series (cumecs) was log-transformed ($\log(1 + x)$) to reduce skewness and stabilize variance. Model predictions were inverse-transformed ($\exp(x) - 1$) for evaluation on the original scale.
- **Scaling:** All predictor variables were normalized to the $[0, 1]$ range using Min–Max scaling. Scaling parameters were estimated from the training split only and consistently applied to validation and test splits to prevent information leakage.

• **Sequence Structuring:** The time series was reformatted into a sequence-to-sequence structure, with a 30-day look-back window used to predict a 5-day inflow horizon. After accounting for this windowing procedure, the final dataset comprised 1770 daily samples, representing the effective number of training–testing instances available for modeling. This preprocessing framework produced a stable, fully numerical dataset suitable for training advanced hybrid ML–DL models for multi-step inflow forecasting. The total feature count was 162 predictors in this study, based on rainfall grid cells and discharge series extracted from the CSV dataset.

III. METHODOLOGY

The proposed framework is a two-stage stacked deep learning–machine learning ensemble for multi-step inflow forecasting. It is motivated by the hydrological challenges of the Indian subcontinent, where rain-fed rivers experience sharp, climate-driven variability. Extreme monsoon floods are among the most devastating natural hazards, and reliable short- to medium-term forecasts are critical for basin management. Our system is explicitly designed to capture both short-term storm events and long-term monsoon-scale dependencies, ensuring robustness under climate non-stationarity.

A. Problem Setup

We address the task of forecasting river inflows in the Mahanadi basin up to five days ahead, framed as a multi-step sequence-to-sequence learning problem. The dataset comprises:

- **Rainfall:** Daily basin-aggregated rainfall from IMD gridded records.
- **Discharge:** Inflows from 11 Central Water Commission (CWC) gauging stations.

In total, 162 predictors are used. The target variable is the basin mean inflow discharge (cumecs). Each training sample is constructed using a 30-day input window and a 5-day forecast horizon, producing a vector output:

$$\mathbf{y} = [y_{t+1}, y_{t+2}, \dots, y_{t+5}]$$

Preprocessing includes:

- **Variance stabilization:** log-transform $\log(1 + x)$.
- **Scaling:** Min–Max scaling to $[0, 1]$.
- **Imputation:** Missing or zero values replaced with a small constant (0.01).

Approximately 1,770 effective daily samples were available after windowing. A held-out independent test set comprising the final 454 chronological samples was reserved for final evaluation.

B. Stage 1: Deep Learning Ensemble

The first stage is a multi-branch deep learning ensemble, where three heterogeneous architectures are trained end-to-end with mean squared error (MSE) loss. Each branch outputs a 5-day forecast vector, and their concatenation forms the high-level feature representation for downstream fusion.

1) *Deep LSTM with Multi-Head Attention:* Two stacked LSTM layers capture sequential dependencies in rainfall–runoff dynamics. An LSTM cell evolves as:

$$\begin{aligned} i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i), \\ f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f), \\ o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o), \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tanh(W_c x_t + U_c h_{t-1} + b_c), \\ h_t &= o_t \odot \tanh(c_t), \end{aligned}$$

where x_t is the input, h_t the hidden state, and c_t the memory cell. Between the two LSTM layers, an 8-head self-attention mechanism enhances feature saliency, learning rainfall–runoff lags that strongly influence inflow:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V.$$

Residual connections and layer normalization stabilize training.

2) *Patch Time Series Transformer (PatchTST-inspired branch):* The 30-day sequence is compressed via a dense layer and treated as a token for attention. While inspired by PatchTST, no Conv1D patch extraction is performed. Instead, this branch implements dense compression + attention to approximate patch-level representation.

3) *TFT-Inspired Hybrid Branch:* The third branch is inspired by the Temporal Fusion Transformer (TFT). It combines:

- Stacked LSTMs for local recurrence, and
- Multi-head attention for long-range dependencies.

Variable selection networks and gating components from the full TFT are not implemented; the branch focuses on combining recurrence (stacked LSTMs) with attention. The outputs of all three branches are concatenated and projected through a dense layer to yield a fused 5-day forecast vector. In addition to producing fused forecasts, the raw horizon-wise outputs from the three branches (5 horizons $\times$ 3 branches = 15 features) are retained as DL meta-features for second-stage modeling.

C. Stage 2: Boosted Meta-Learning and Blending

To refine base forecasts, a second-stage ensemble employs horizon-specific gradient-boosted decision trees (XGBoost).

- **Meta-features:** The second-stage meta-learner receives a 15-dimensional DL feature vector (3 DL branches $\times$ 5 forecast horizons). These capture diverse temporal dynamics across models and horizons. In addition, out-of-fold (OOF) predictions from auxiliary ML learners (XGBoost, LightGBM, Random Forest) are included per horizon, along with ensemble statistics (mean, std, median). For DL, no OOF procedure is applied; instead, train/validation predictions are used. For ML learners, strict OOF predictions are generated via chronological TimeSeriesSplit.
- **Peak-aware weighting:** Samples above the 90th percentile and later horizons are upweighted with day-dependent weights $[1.0, 1.0, 1.2, 1.4, 1.6]$, improving flood-peak capture.
- **Blending:** Final forecasts combine boosted meta-model outputs with the arithmetic mean of DL base learners:

$$\tilde{y}_d = 0.6 \hat{y}_d^{\text{boost}} + 0.4 \frac{1}{3} \sum_{k=1}^3 \hat{y}_d^{(k)},$$

where $\hat{y}_d^{(k)}$ is the prediction from branch k for day d .

OOF Clarification: Out-of-fold (OOF) predictions are generated for the ML base learners (XGBoost, LightGBM, Random Forest) to ensure unbiased meta-features. For DL stacked outputs, direct model predictions on train/validation splits are used (no DL OOF generation).

D. Architectural Reasoning

1) *Why Multi-Head Attention?*: The Mahanadi basin exhibits strong spatio-temporal heterogeneity, where rainfall in one sub-region can trigger flow responses at distant downstream gauges after variable lags. Multi-head attention is particularly well-suited to capture these complexities for several reasons:

- **Capturing spatial dependencies:** Rainfall–runoff processes are inherently non-local. Multi-head attention allows each head to specialize in distinct relationships. By aggregating these diverse perspectives, the model constructs a rich, multi-faceted representation of basin-scale hydrodynamics.
- **Self-learning attention scores:** Unlike static weighting schemes, attention scores are optimized end-to-end during training. This self-learning capability allows the architecture to tailor its internal focus to the specific hydro-climatic patterns of the basin.
- **Temporal causality:** Causality is maintained by design: the sequence-to-sequence setup and attention layers only use past and present inputs to inform predictions. This ensures that forecasts remain physically consistent.
- **Complementarity with LSTMs:** Standard LSTMs form the backbone of the architecture, capturing sequential memory. Multi-head attention augments them by modeling long-range cross-feature interactions. This dual mechanism enhances both local sensitivity and global context-awareness.

2) *Short- vs. Long-Term Learning:* The proposed stacked ensemble is explicitly designed to balance short- and long-term hydrological dynamics:

- **Short-term learning (ML residual correction):** The gradient-boosted meta-learner operates as a corrective layer, identifying systematic biases in the deep learning base forecasts.
- **Long-term learning (DL sequence models):** LSTM and Transformer-inspired branches are adept at capturing seasonal and monsoon-scale dependencies.
- **Ensemble diversity:** Integrating three heterogeneous backbones reduces shared biases and enhances predictive robustness.
- **Hydro-climatic suitability:** The ensemble is explicitly tailored to the Indian monsoon system, enabling it to model high-frequency storm bursts, low-flow dry spells, and intra-seasonal rainfall variability.

E. Training and Evaluation

- **Implementation:** Deep learning in TensorFlow/Keras; ML models in XGBoost, LightGBM, scikit-learn.
- **Optimization:** Adam optimizer ($lr = 3 \times 10^{-4}$), batch size 32, 200 epochs with ReduceLROnPlateau and EarlyStopping.
- **Testing:** Independent held-out test set covering recent extremes.

1) *Metrics:* Hydrological forecasting skill was evaluated using both standard efficiency/error criteria and flood-peak metrics:

- **Nash–Sutcliffe Efficiency (NSE):**

$$NSE = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}.$$

- **Kling–Gupta Efficiency (KGE):**

$$KGE = 1 - \sqrt{(r - 1)^2 + (\beta - 1)^2 + (\gamma - 1)^2},$$

where r = correlation, β = bias ratio, γ = variability ratio.

- **Root Mean Square Error (RMSE):**

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}.$$

- **Mean Absolute Error (MAE):**

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|.$$

While hydrological forecasting skill was primarily assessed using NSE, KGE, RMSE, and MAE, additional peak-focused metrics were computed:

- **Total Peak Error (TPE):** Sum of absolute errors for flows above the 90th percentile of observed discharge.
- **Peak Evolution Error (TPE_Evol):** Normalized error in capturing the temporal evolution of peaks (first differences). These peak metrics were treated as supplementary indicators of flood-peak representation, while the main evaluation emphasizes NSE, RMSE, MAE, and KGE. Random seed was fixed at 42 for reproducibility.

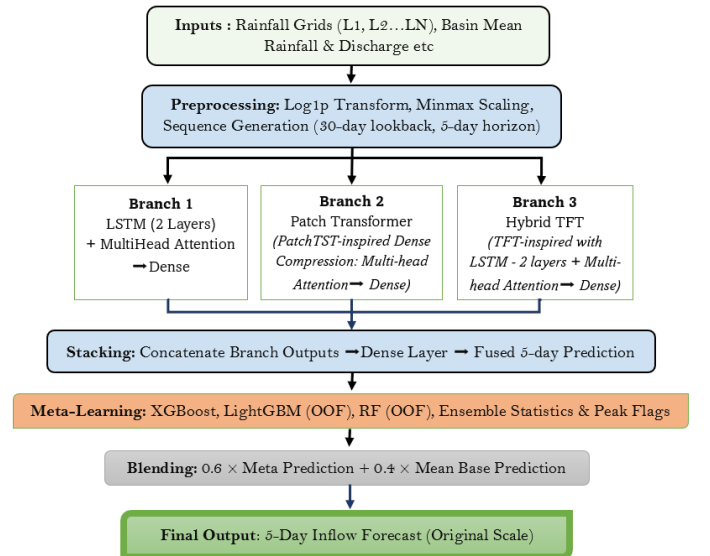


Fig.: Block diagram of the hybrid forecasting framework showing preprocessing, deep learning branches, meta-learning stacking, and final 5-day inflow prediction.

I. RESULTS AND DISCUSSION

A. Experimental Setup and Evaluation Metrics

The performance of the proposed Boosting+Stacking hybrid framework and the baseline Stacked DL-only model was evaluated on a held-out test set comprising the final 454 samples of the time series. To ensure reproducibility, all experiments were conducted with fixed random seeds (42). Data splitting and scaling were performed chronologically, and scalers were fit exclusively on the training set to prevent data leakage.

Model performance was quantified for each day of the 5-day forecast horizon using widely adopted hydrological skill metrics:

- **Nash–Sutcliffe Efficiency (NSE):** Evaluates predictive skill by comparing model residual variance to the variance of observed flows. NSE values range from $-\infty$ to 1, with 1 denoting a perfect fit. Forecasts are typically considered skillful if $NSE \geq 0.70$.
- **Root Mean Square Error (RMSE) and Mean Absolute Error (MAE):** Represent average forecast errors in the original inflow units (cumecs). Lower values indicate higher accuracy and reduced bias.
- **Kling–Gupta Efficiency (KGE):** A composite indicator that decomposes error into correlation, variability bias, and mean bias. KGE values close to 1 indicate better performance, with values above 0.41 reflecting skill relative to the mean flow benchmark.

Peak-specific indices such as TPE and TPE_Evol were computed to complement the main evaluation. However, the primary emphasis remains on NSE, RMSE, MAE, and KGE, which capture overall model reliability. Peak-oriented measures are acknowledged as supplementary and promising for future systematic use.

B. Quantitative Results

The comparative performance across the 5-day forecast horizon is summarized in Tables I and II.

TABLE I
5-DAY FORECAST PERFORMANCE OF THE PROPOSED
BOOSTING+STACKING MODEL

Day	NSE	RMSE (cumecs)	MAE (cumecs)	KGE
1	0.961	751.05	412.56	0.940
2	0.924	1045.48	509.69	0.922
3	0.840	1512.59	641.14	0.880
4	0.755	1870.52	663.42	0.807
5	0.712	2031.03	766.05	0.758

TABLE II
5-DAY FORECAST PERFORMANCE OF THE STACKED DL-ONLY MODEL

Day	NSE	RMSE (cumecs)	MAE (cumecs)	KGE
1	0.857	1430.82	668.18	0.698
2	0.827	1573.53	692.52	0.657
3	0.769	1815.71	762.68	0.648
4	0.678	2144.41	871.01	0.578
5	0.633	2290.92	917.80	0.559

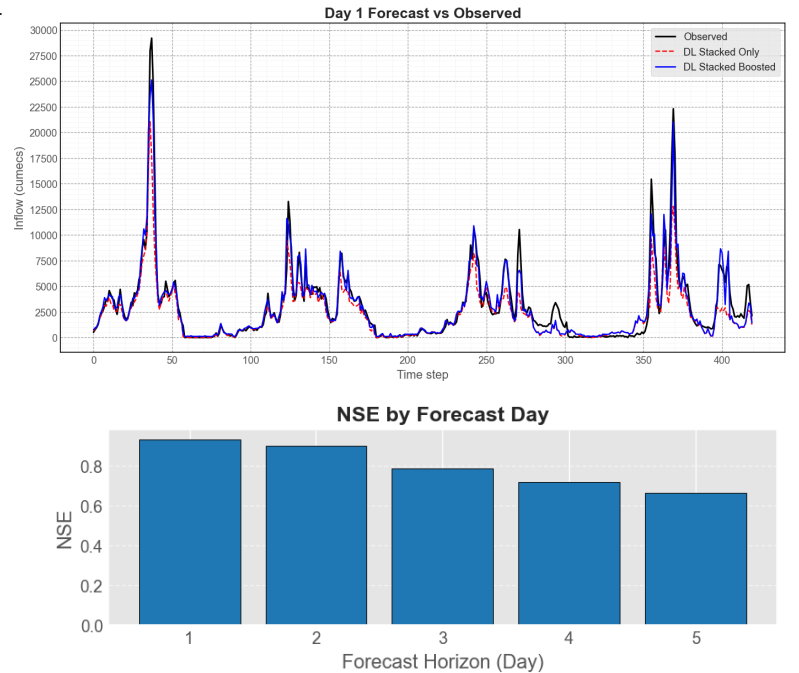
C. Discussion of Results

The results demonstrate the consistent superiority of the Boosting+Stacking framework over the Stacked DL-only baseline across all horizons and metrics.

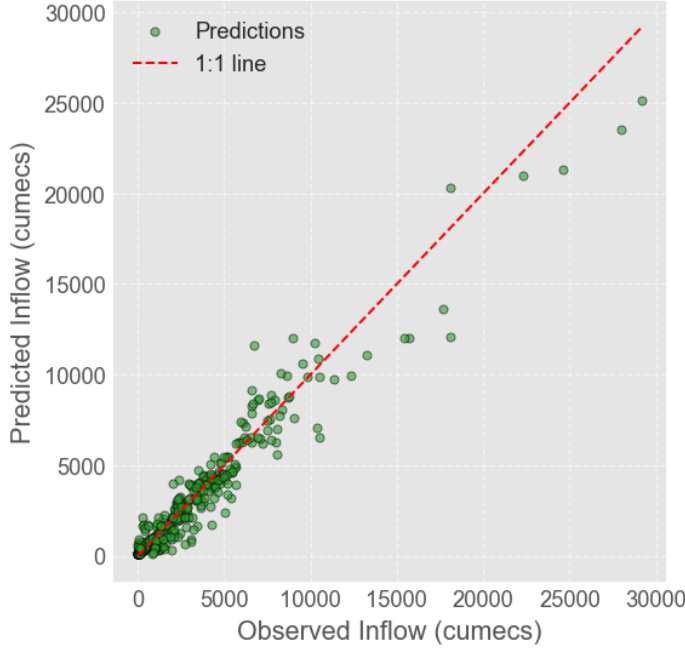
- **Short-term forecasts (Day-1 and Day-2):** The hybrid approach shows striking improvements. For Day-1, NSE improves from 0.857 (DL-only) to 0.961, corresponding to a $>70\%$ reduction in unexplained variance. RMSE decreases by 47.5% (1430.82 $\rightarrow$ 751.05 cumecs) and MAE by 38.3% (668.18 $\rightarrow$ 412.56 cumecs). KGE also improves markedly (0.698 $\rightarrow$ 0.940), reflecting stronger correlation with observed flows and reduced bias. These gains persist through Day-2 (NSE = 0.924, KGE = 0.922), confirming the model’s reliability for early flood warning applications.
- **Medium-range forecasts (Day-3 to Day-5):** Performance naturally declines as lead time increases, reflecting compounding uncertainty. Nevertheless, the hybrid model maintains acceptable predictive skill ($NSE \geq 0.70$) through Day-5, whereas the DL-only ensemble drops below this threshold by Day-4. This highlights the hybrid model’s greater robustness under long-range prediction scenarios.

The observed improvements can be attributed to the hierarchical learning structure. Deep learning branches capture multi-scale temporal dependencies, while the XGBoost meta-learner acts as a corrective mechanism, reducing systematic biases and learning optimal non-linear combinations of base forecasts. This synergy enhances both stability and accuracy, particularly during flood peaks, where errors can have the greatest operational impact.

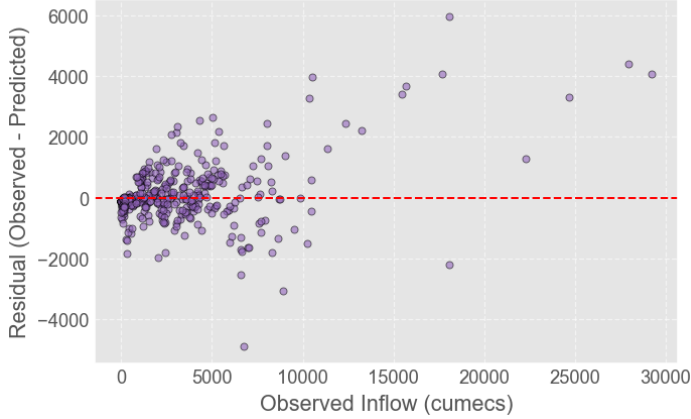
Despite these advances, a gradual performance decline after Day-3 underscores the challenge of multi-step inflow forecasting. As uncertainty compounds, model skill diminishes. Addressing this limitation will likely require additional exogenous predictors (e.g., soil moisture indices, satellite-derived rainfall estimates, or quantitative precipitation forecasts).



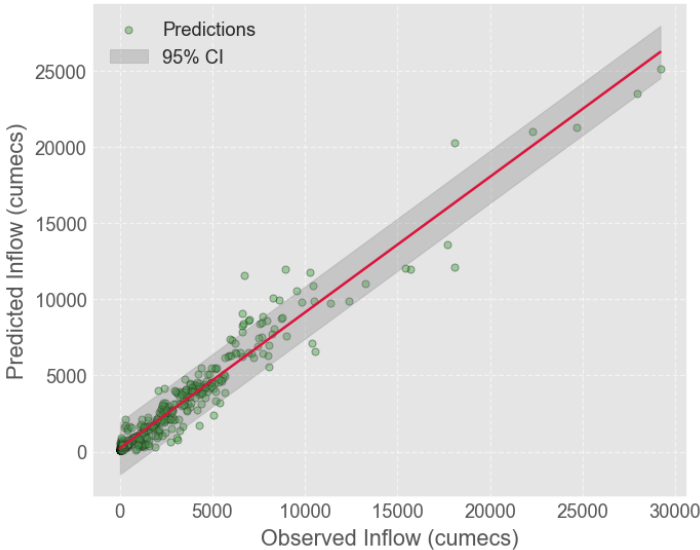
Day 1: Observed vs Predicted



Day 1: Residuals vs Observed



Day 1: Regression with 95% CI



II. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper presented and validated a hybrid deep learning + machine learning ensemble framework for short-term flood forecasting in the Mahanadi Basin. The framework fuses multiple sequence-based deep learning models (LSTM, PatchTST, TFT-like architectures) with an XGBoost meta-learner for decision-level fusion.

The deep learning branches capture both short-term and long-term rainfall-runoff dynamics, while the boosted meta-learner refines outputs by correcting systematic residual errors. The use of multi-head attention within the deep learning modules was crucial for capturing spatial dependencies across the basin and ensuring strict temporal causality in the forecasts.

Key outcomes include:

- **Short-term accuracy:** NSE > 0.90 sustained through Day-2 forecasts.
- **Medium-range robustness:** Skillful performance (NSE ≥ 0.70) maintained across the full 5-day horizon, even under increasing forecast uncertainty.
- **Error reduction:** Substantial decreases in RMSE and MAE, alongside consistently higher KGE scores, particularly in reproducing high-flow dynamics that are critical for flood risk management.
- **Peak flow representation:** Although overall skill improved, peak-focused supplementary metrics (TPE, TPE_Evol) remain areas for further enhancement.

These findings underscore the value of a hierarchical ensemble design, where deep learning provides powerful temporal feature extraction and boosted trees deliver non-linear error correction. The resulting synergy yields forecasts that are not only more accurate but also more operationally reliable for early flood warning in vulnerable river basins.

B. Limitations and Future Work

While the proposed framework demonstrates substantial improvements, several limitations should be acknowledged:

- **Computational complexity:** Training and inference require significant resources, which may constrain real-time deployment.
- **Data dependence:** Reliable performance depends on the availability of long, continuous, high-quality hydrometeorological datasets.
- **Peak flow representation:** Although overall skill improved, peak-focused metrics (e.g., TPE, TPE_Evol) remain an area where further enhancement is necessary.

Future work could focus on:

- **Integration of exogenous predictors** such as satellite rainfall, soil moisture, and QPFs to enhance medium-range skill.
- **Advanced spatial modeling** using Graph Neural Networks (GNNs) to explicitly represent basin topology and rainfall-runoff connectivity.
- **Probabilistic forecasting frameworks** to provide prediction intervals and quantify uncertainty, enabling risk-informed flood management.